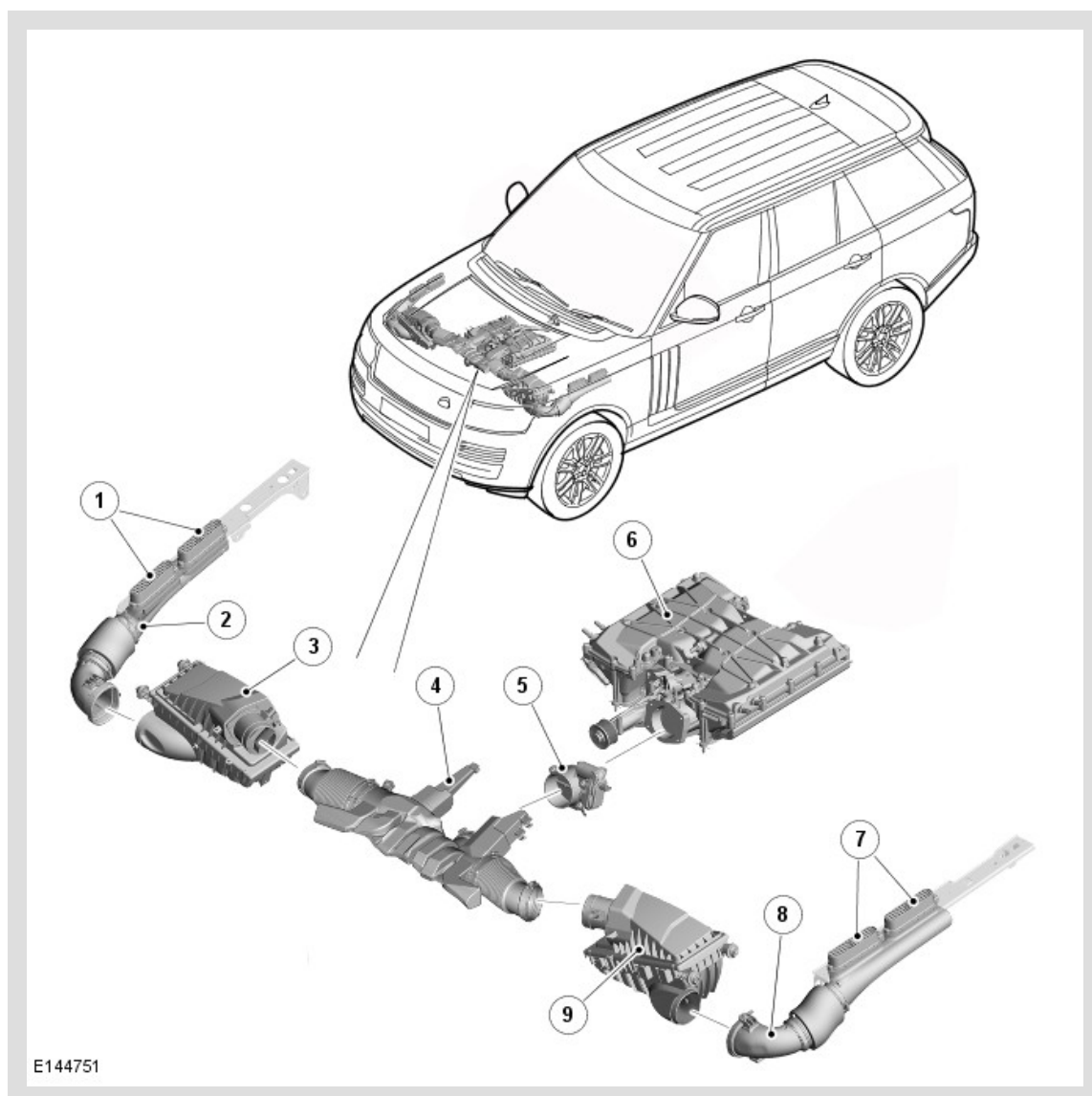


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2014.0 RANGE ROVER (LG), 303-12

INTAKE AIR DISTRIBUTION AND FILTERING - V8 S/C 5.0L PETROL

COMPONENT LOCATION



ITEM	DESCRIPTION
1	Right intake grilles
2	Right dirty air duct
3	Right air cleaner
4	Clean air ducts
5	Electric throttle
6	Supercharger and intake manifolds

7	Left intake grilles
8	Left dirty air duct
9	Left air cleaner

OVERVIEW

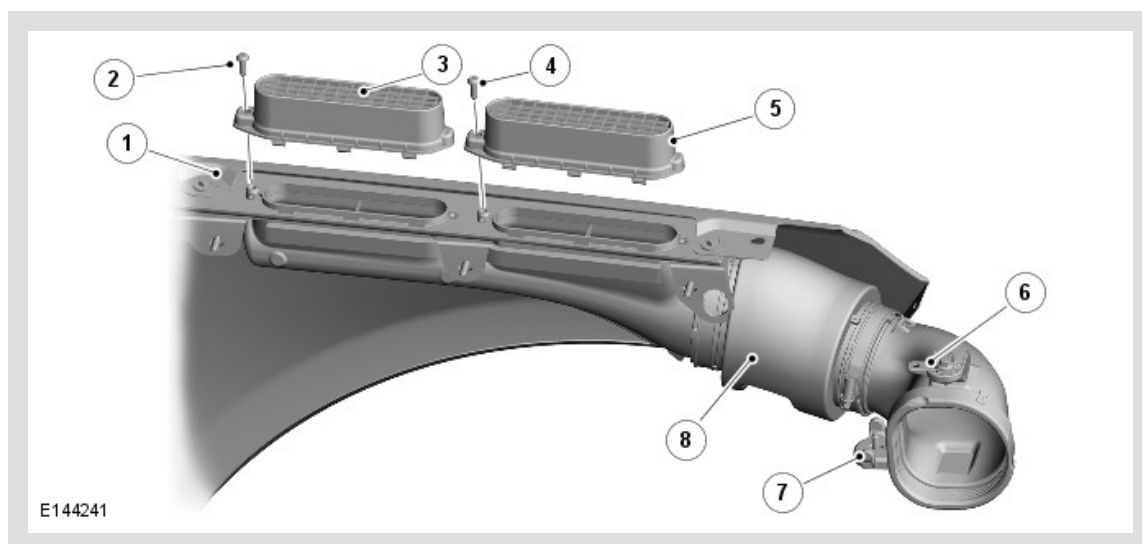
The intake air distribution and filtering system comprises:

- Two dirty air ducts.
- Two air cleaners.
- Clean air ducts.
- An electric throttle.
- A SC (supercharger) and intake manifolds.

Air flows through the dirty air ducts, air cleaners, clean air ducts and electric throttle to the SC . The SC compresses the air, which is then directed through charge air coolers in the intake manifolds and into the cylinders.

DESCRIPTION

DIRTY AIR DUCTS



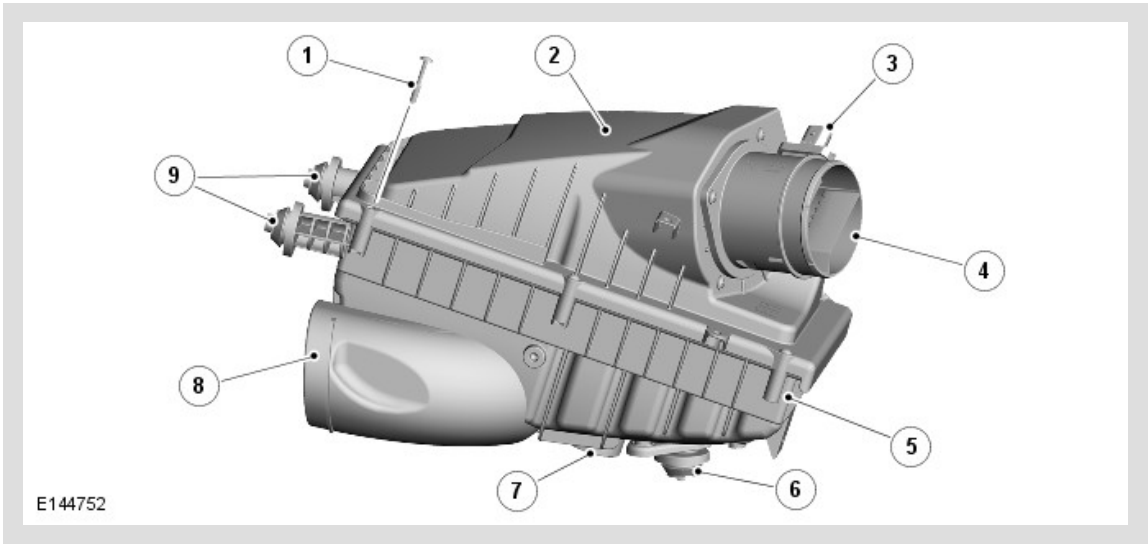
ITEM	DESCRIPTION
1	Front fender support
2	Screw
3	Intake grille
4	Screw
5	Intake grille
6	Front upper bracket
7	Front lower bracket
8	Non-porous sleeve

The dirty air ducts are installed in the front fenders to transfer ambient air from under the edges of the hood to the air cleaners.

Each dirty air duct locates in the intake of the related air cleaner and is attached to the front inner fender with two brackets. The intake of each dirty air duct is attached to two intake grilles on top of the front fender support. When the hood is closed, the intake grilles locate in guides attached to the hood inner panel.

A porous section, covered by a non-porous sleeve to prevent moisture ingress, is incorporated into each dirty air duct to reduce induction noise.

AIR CLEANERS



ITEM	DESCRIPTION
1	Screw (6 off)
2	Lid
3	Mass air flow and temperature sensor
4	Air outlet
5	Base
6	Isolator
7	Drain valve
8	Air intake
9	Isolators

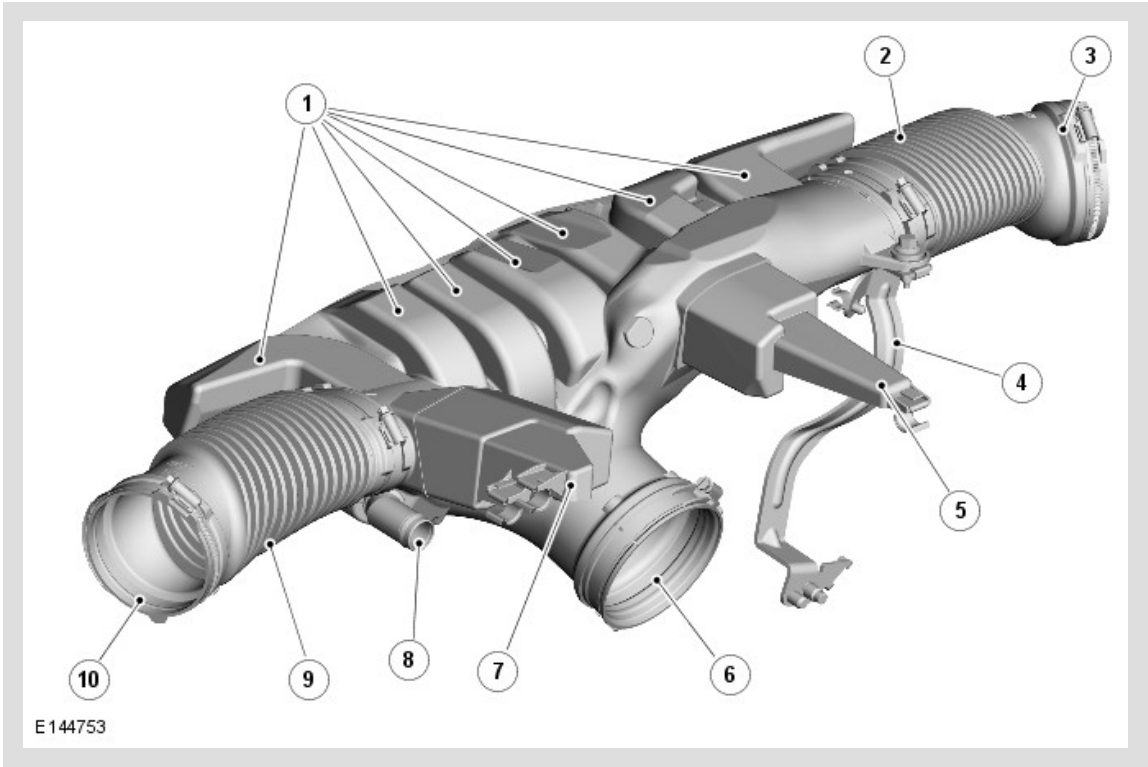
An air cleaner is located in each front corner of the engine compartment. Three spigots fitted with isolator bushes locate each air cleaner in holes in the related front inner fender and front suspension housing.

Each air cleaner consists of a filter element installed in a base and enclosed with a lid secured by six screws. The filter element is a pleated, paper type with an integral seal.

Air intake and outlet connections are incorporated into the base and lid respectively. The bottom of the base incorporates a drain valve to prevent the accumulation of water in the air cleaner.

The air outlet connection of each air cleaner incorporates a MAFT (mass air flow and temperature) sensor. The sensors are connected to the ECM (engine control module) . For additional information, refer to: Electronic Engine Controls (303-14D, Description and Operation).

CLEAN AIR DUCTS



ITEM	DESCRIPTION
1	Resonators
2	Right convolute
3	Right air intake
4	Support bracket
5	Resonator
6	Air outlet
7	Resonator
8	Full load breather connector stub
9	Left convolute

10	Left air intake
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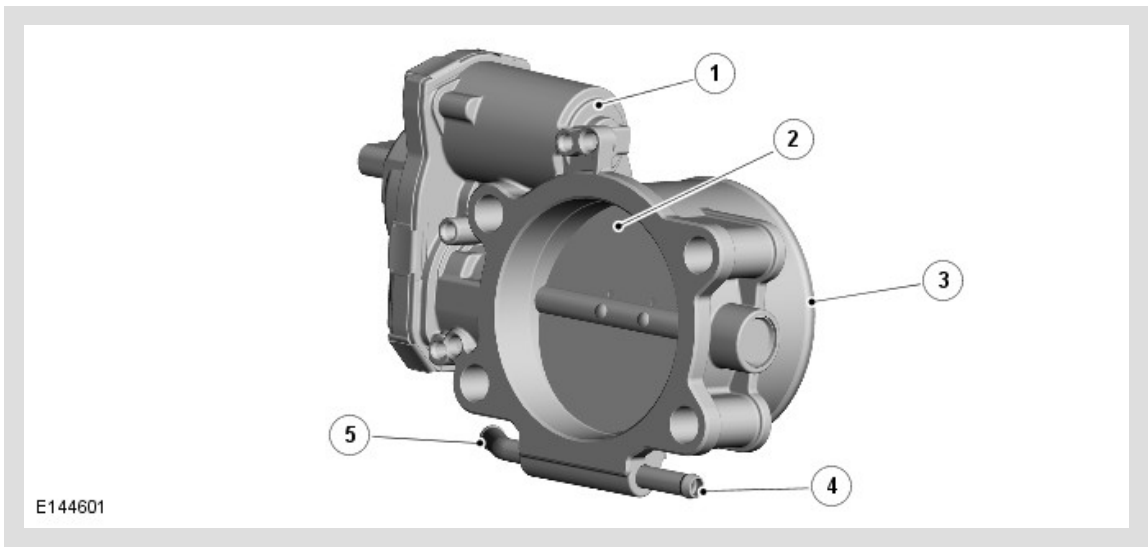
The clean air ducts transfers air from the air cleaner outlets to the electric throttle. Two convolute sections in the clean air ducts accommodate the relative movement between the engine mounted electric throttle and the body mounted air cleaners.

The clean air duct also incorporates the following:

- Resonators, to reduce air induction noise.
- A connector stub for the engine full load breather pipe.

The clean air duct is supported by a bracket attached to the right cylinder head.

ELECTRIC THROTTLE



ITEM	DESCRIPTION
1	Electric motor
3	Air intake
3	Air outlet
4	Coolant feed connection
5	Coolant return connection

The electric throttle regulates the air flow into the SC .

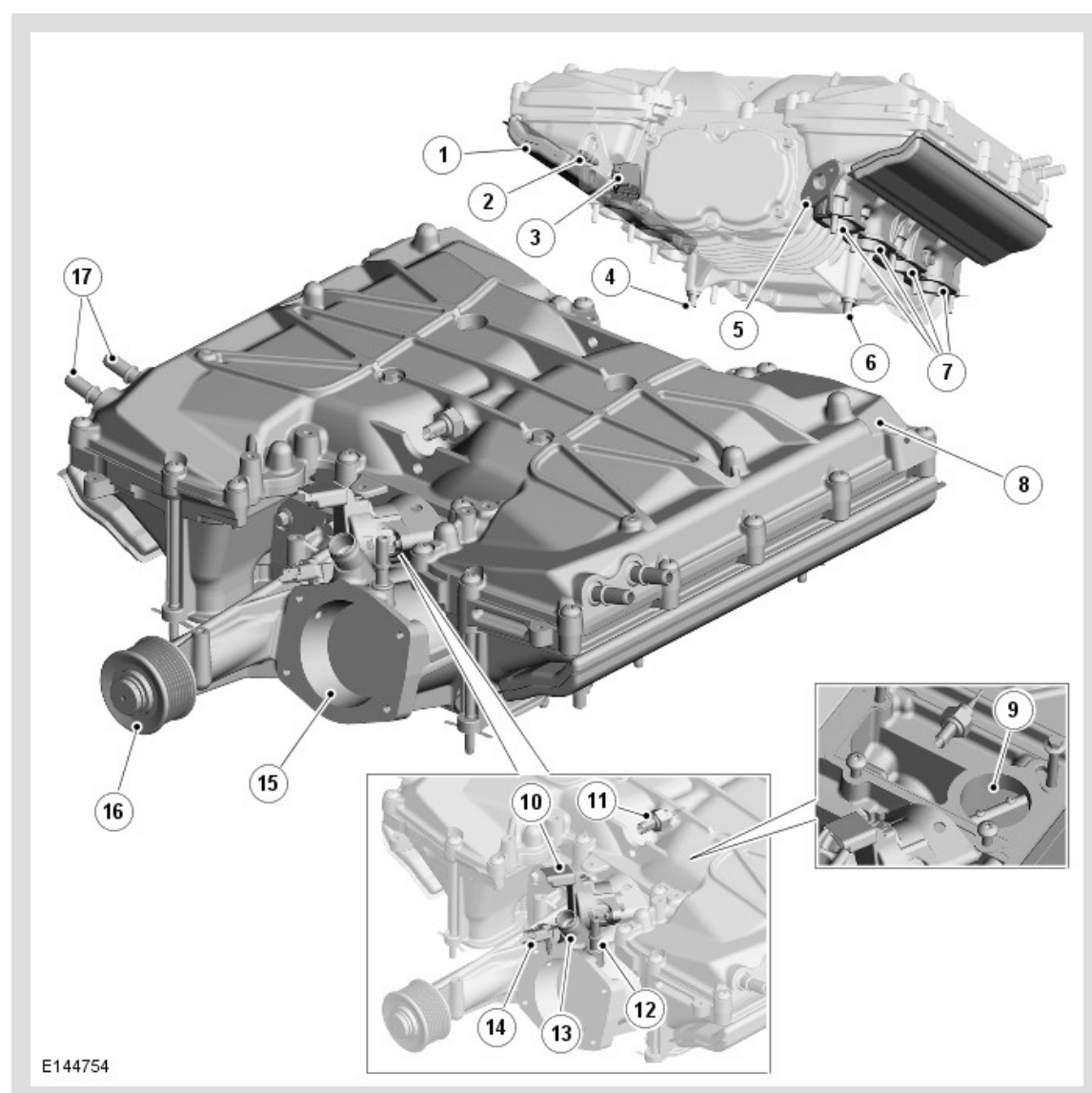
The electric throttle is installed at the front of the engine, between the cylinder heads. Four bolts attach the electric throttle to the SC .

The throttle plate is operated by a DC (direct current) electric motor attached to the throttle body. The motor is controlled by the ECM and is constantly adjusted in response to driver inputs with the throttle pedal to precisely control the amount of air allowed into the intake manifold.

For additional information, refer to: Electronic Engine Controls (303-14D, Description and Operation).

To prevent icing, coolant is circulated across the throttle body through coolant feed and return connections.

SUPERCHARGER AND INTAKE MANIFOLDS



ITEM	DESCRIPTION
1	NVH pad
2	Vacuum connector stub
3	Temperature and manifold absolute pressure sensor
4	Dowel
5	Symposer intake pipe connection
6	Dowel
7	Intake manifold outlet ports
8	Charge air cooler tank top
9	Bypass valve
10	Bypass valve actuator
11	Supercharger outlet temperature sensor
12	Evaporative emissions connector stub
13	Part load breather connector stub
14	Manifold absolute pressure sensor
15	Supercharger intake port
16	Pulley
17	Coolant intake and outlet connections for charge air cooler

The SC increases the pressure, and thus mass, of the air supplied to the engine, to increase the engine's power output. Two separate intake manifolds direct air from the SC to the cylinder intake ports.

The intake manifolds are attached to their related cylinder heads and the sides of the SC . Two dowels locate the SC in position on the cylinder block. A charge air cooler tank top is installed on top of the SC and intake manifolds to form the air duct from the SC outlet to the intake manifolds. A charge air cooler is installed in each intake manifold.

A NVH (noise, vibration and harshness) pad is attached to the side of each intake manifold. The rear of the right intake manifold incorporates a connection port for the symposer intake pipe. The rear of the left intake manifold incorporates:

- A connector stub for the brake vacuum system.
- A TMAP (temperature and manifold absolute pressure) sensor.

The SC is a Roots blower with high angle helix rotors driven at 2.1 times engine speed by the secondary belt of the accessory drive.

The two rotors of the SC are contained in a housing. The ends of the rotors are supported in bearings in the front cover and the bearing plate. A rear cover seals the bearing plate. A pulley transfers power from the accessory drive to the shaft of one of the rotors.

A DC electric actuator on the front cover is attached to a bypass valve in the housing. The bypass valve allows air to bypass the rotors, to control the outlet pressure of the SC . Operation of the actuator is controlled by a PWM (pulse width modulation) signal from the ECM . A Hall effect position sensor in the actuator returns a 0.5 to 4.5 V signal to the ECM . This allows the ECM to identify the position of the bypass valve for closed-loop control.

The front cover also incorporates:

- The SC air intake and mounting face for the electric throttle.
- A connector stub for the part load breather.
- A MAP (manifold absolute pressure) sensor.
- A connector stub for a hose from the EVAP (evaporative emission) canister purge valve.

OPERATION

SUPERCHARGER

At closed or partially open throttle positions, the bypass valve is fully open, allowing intake air to bypass the SC . This results in little or no pressure increase across the SC . Progressive opening of the throttle reduces the depression downstream of the electric throttle. This is sensed by the ECM from the MAP sensor input. The ECM then signals the bypass valve actuator to progressively close the bypass valve. As the bypass valve closes there is a corresponding increase in the outlet pressure from the SC , which increases engine power output.